

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Assessment Tool Weightage: 40%

QUESTIONS: [08]

Total Marks:25

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Q5. Regularized CNN vs Non-Regularized CNN

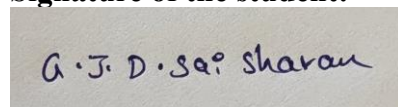
An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization** and a **CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Code of Conduct:

I G J D Sai Sharan/192425152 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature reads "G. J. D. Sai Sharan".

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Q1. CNN vs Fully Connected Neural Network

A **Convolutional Neural Network (CNN)** is more suitable for X-ray image classification than a traditional **Fully Connected Neural Network (FCNN)** because CNNs are specifically designed to preserve and learn spatial relationships in images.

Comparison

Feature	CNN	Fully Connected Neural Network
Architecture	Convolution, activation, pooling, and fully connected layers	Mainly fully connected layers
Motivation	Designed for image and spatial data	Designed for general numerical/vector data
Input handling	Maintains image height, width and channels	Image is usually flattened into a 1D vector
Spatial feature extraction	Automatically extracts local spatial features	Spatial relationships are largely lost after flattening
Parameters	Fewer due to local connectivity and parameter sharing	Very large number of parameters
Computational complexity	Lower for image feature extraction	Very high for large images
Memory requirement	Relatively lower	Relatively higher
Translation handling	More robust to small shifts in objects	Less effective
Image suitability	Excellent	Poor for high-resolution images

Role of Convolutional Layers

Convolutional layers apply learnable **filters/kernels** to the image. Each filter detects a particular pattern.

For example:

- First layers → edges and simple textures
- Middle layers → shapes and patterns
- Deeper layers → complex structures and disease-related features

The same filter is applied across different locations of the image, which is called **parameter sharing**.

Role of Filters

Filters are small matrices such as 3×3 or 5×5 . During training, their weights are learned automatically.

A filter may learn to detect:

- Horizontal edges
- Vertical edges
- Curves
- Textures
- Shapes
- Medical abnormalities

Role of Pooling

Pooling reduces the spatial dimensions of feature maps.

Common types:

- **Max pooling** → selects the maximum value
- **Average pooling** → calculates the average value

Pooling reduces computation and provides some robustness to small translations.

Why CNN is better for X-ray classification

An X-ray contains important spatial information. For example, the position and shape of abnormal regions are important for detecting diseases. CNNs preserve this information while automatically learning useful features.

A fully connected network requires flattening the image, which produces a huge number of parameters.

Conclusion: CNNs are preferred because they provide **local connectivity, parameter sharing, automatic feature extraction, reduced parameters, lower computational requirements, and better image classification performance.**

Q2. Different CNN Layers and Filters

CNNs use different layers to progressively transform a raw image into meaningful features and finally classify it.

Comparison of CNN Layers

Layer	Main Function	Spatial Information	Computational Requirement	Main Purpose
Convolution	Extracts features	Preserves spatial relationships	Moderate	Feature extraction

Pooling	Reduces feature-map size	Reduces spatial resolution	Low	Dimensionality reduction
Activation	Adds non-linearity	Maintains dimensions generally	Low	Learn complex patterns
Fully Connected	Combines extracted features	Spatial structure mostly removed	High	Final classification

1. Convolutional Layer

The convolutional layer applies filters to the input image.

It detects features such as:

- Edges
- Lines
- Corners
- Textures
- Shapes

For industrial defect detection, convolutional layers can detect scratches, cracks, holes, dents, and irregular patterns.

2. Pooling Layer

Pooling reduces the size of feature maps.

For example:

$$28 \times 28 \rightarrow 14 \times 14$$

Advantages:

- Reduces computation
- Reduces memory requirements
- Reduces sensitivity to small changes
- Helps control overfitting

3. Activation Layer

A common activation function is **ReLU**:

$$\text{ReLU}(x) = \max(0, x)$$

It introduces non-linearity, allowing the CNN to learn complex relationships.

4. Fully Connected Layer

Fully connected layers combine the learned high-level features and produce the final classification.

For example:

Feature extraction → Defect features → Classification

The final layer can classify products as:

Defective

Non-defective

Early vs Deeper Filters

Early Filters	Deeper Filters
Learn simple features	Learn complex features
Edges	Object parts
Lines	Shapes
Corners	Textures
Basic patterns	Defect patterns
Low-level features	High-level semantic features

For example:

Image

↓

Edges

↓

Textures

↓

Shapes

↓

Crack/Scratch patterns

↓

Defective / Non-defective

Conclusion

The layers work together as a hierarchy. **Convolution extracts features, activation introduces non-linearity, pooling reduces dimensionality, and fully connected layers perform classification.** Early filters detect simple patterns while deeper filters combine them into complex defect representations. Therefore, CNNs are highly effective for automated industrial defect detection.

Q3. Parameter Sharing vs Independent Weights

Parameter sharing is one of the most important characteristics of CNNs. A CNN uses the same convolutional filter across different locations, whereas a fully connected network generally uses separate weights for connections between neurons.

Comparison

Feature	CNN – Parameter Sharing	Fully Connected – Independent Weights
Weights	Same filter reused across image	Different weights for different connections
Number of parameters	Much lower	Very high
Memory	Lower	Higher
Computation	More efficient for image features	Computationally expensive
Feature detection	Same feature can be detected anywhere	Position-specific learning
Translation handling	More robust to shifts	Less robust
Learning efficiency	High	Lower for image data
Scalability	Suitable for large images	Difficult for high-resolution images

Example

Suppose an image has:

$$224 \times 224 \times 3$$

The input contains:

$$224 \times 224 \times 3 = 150,528$$

values.

A fully connected layer with 1,000 neurons would require approximately:

$$150,528 \times 1,000$$

weights, which is more than **150 million parameters**, before considering biases.

A CNN may use a small 3×3 filter. That filter is reused across the entire image.

Parameter Sharing

Suppose a CNN has a 3×3 filter:

Filter:

[w1 w2 w3
w4 w5 w6
w7 w8 w9]

Only a small number of weights are learned, and the same weights are applied at different image locations.

Therefore, if the same edge appears in the top-left and bottom-right regions, the same filter can detect it.

Advantages

1. Fewer parameters

Parameter sharing dramatically reduces the number of trainable weights.

2. Lower memory usage

Fewer parameters require less memory.

3. Faster learning

The same filter learns from many different locations.

4. Better feature detection

The network can recognize a feature regardless of where it occurs.

5. Translation-related robustness

If an object moves slightly within an image, convolution can still detect its features.

Conclusion

Parameter sharing makes CNNs **more memory-efficient, computationally efficient, and effective at learning reusable visual features**. This is particularly important when processing thousands of high-resolution surveillance images.

Q4. AlexNet vs ResNet

AlexNet and **ResNet** are important CNN architectures, but ResNet is generally more suitable for complex modern image recognition tasks.

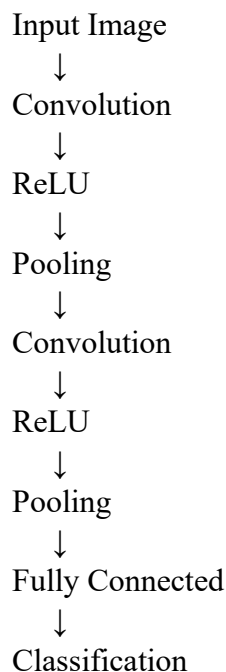
Comparison

Feature	AlexNet	ResNet
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Introduction	2012	2015
Depth	8 learnable layers	18, 34, 50, 101, 152+ layers
Architecture	Conventional sequential CNN	CNN with residual/skip connections
Convolution filters	Large early filters such as 11×11	Mostly smaller 3×3 filters
Parameters	About 60 million	Depends on variant; e.g., ResNet-50 about 25.6 million
Feature extraction	Good	Excellent for deep feature hierarchies
Training difficulty	Moderate	Easier to train very deep networks
Vanishing gradient	More vulnerable	Skip connections help address it
Computational cost	Relatively lower for original architecture	Higher for deeper variants
Performance	Strong for its time	Generally superior on complex tasks
Deep networks	Limited	Designed for very deep networks

AlexNet Architecture

A simplified structure is:



AlexNet was historically important because it demonstrated the effectiveness of deep CNNs for large-scale image classification.

ResNet Architecture

ResNet introduces **residual/skip connections**.

Instead of learning:

$H(x)$

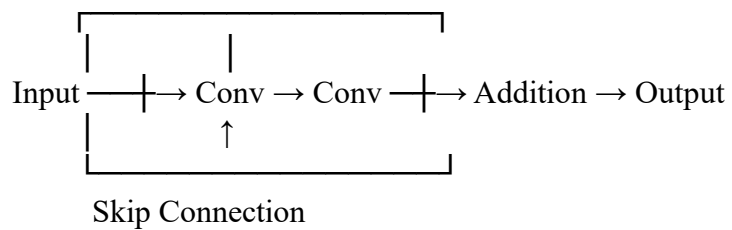
the residual block learns:

$$F(x) = H(x) - x$$

and produces:

$$H(x) = F(x) + x$$

Simplified:



Vanishing Gradient Problem

As networks become deeper, gradients may become extremely small during backpropagation. This makes training difficult.

ResNet's skip connections provide a direct path for information and gradients, making very deep networks easier to optimize.

Feature Extraction

AlexNet learns useful features through successive convolutional layers.

ResNet can build a much deeper hierarchy:

Edges
↓
Textures
↓
Shapes
↓
Object parts
↓
Complex structures
↓
High-level objects

Recommendation

For a **complex image recognition application**, ResNet is the better choice.

Reasons:

1. Supports very deep architectures.
2. Uses residual connections.
3. Reduces the effect of vanishing gradients.
4. Learns richer hierarchical features.
5. Usually provides better classification performance.
6. Can be adapted using transfer learning.

Conclusion: AlexNet is historically significant and computationally simpler, but **ResNet is generally more suitable for complex modern computer vision applications because of its depth, residual learning, better optimization, and stronger feature representation.**

Q5. Regularized CNN vs Non-Regularized CNN

The agricultural CNN has **high training accuracy but poor performance on unseen images**. This is a classic sign of **overfitting**.

Comparison

Feature	Without Regularization	With Regularization
Training accuracy	Very high	Usually slightly lower
Test accuracy	Often poor	Usually better
Overfitting	High	Reduced
Generalization	Poor	Better
Training stability	May be unstable	Batch normalization improves stability
Memory/Computation	Lower	Higher
Data requirements	More prone to overfitting with limited data	Data augmentation effectively increases training diversity
Real-world performance	Poorer	Better

1. Dropout

Dropout randomly disables some neurons during training.

For example:

Before dropout:

● ● ● ● ● ●

After dropout:

● × ● × ● ●

This prevents the network from depending too heavily on particular neurons.

Benefit

- Reduces overfitting
- Improves generalization

2. Data Augmentation

Data augmentation creates modified versions of existing training images.

Examples:

- Rotation
- Flipping
- Cropping
- Scaling
- Translation
- Brightness changes

For leaf disease classification, augmentation can help the model recognize disease symptoms under different orientations and lighting conditions.

Benefit

It effectively increases the diversity of training data without requiring new images to be manually collected.

3. Batch Normalization

Batch normalization normalizes intermediate activations during training.

Benefits include:

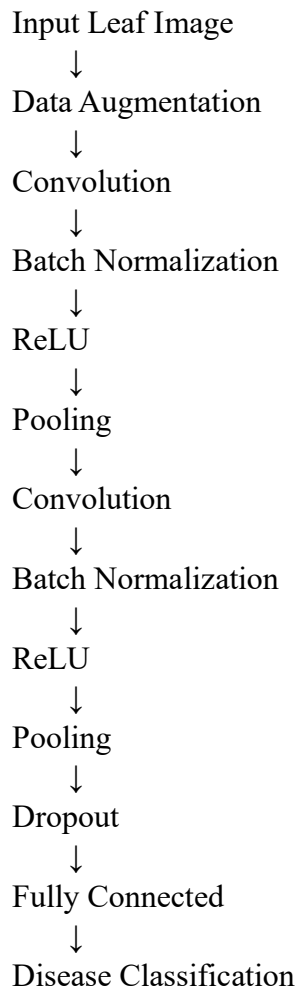
- More stable training
- Faster convergence
- Better gradient flow
- Some regularization effect

It is commonly placed after convolution/linear operations and before or around activation depending on the architecture.

Recommended Approach

For plant disease classification, a combination of **data augmentation + batch normalization + appropriate dropout** is preferable.

A typical architecture could be:



Why this approach is appropriate

Data augmentation improves robustness to real-world variations.

Batch normalization improves training stability and optimization.

Dropout reduces over-dependence on specific neurons and helps control overfitting.

However, excessive dropout or aggressive augmentation can hurt performance, so these should be tuned using validation performance.

Final Conclusion

The original model is **overfitting** because its training accuracy is high while its unseen-image performance is poor. A regularized CNN is more appropriate because it provides **better generalization, improved robustness, and more reliable real-world disease classification**.